Polynomials, Exponentials and Logarithms

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Polynomials

Exercise 0.1 (Reading a polynomial)**.** Let $f(x) = 3x^4 - x + 7$.

1. Write f in the form $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, listing every coefficient a_4, a_3, a_2, a_1, a_0.
2. What is the degree of f, and what is its leading coefficient?

Solution

Solution 0.1.

1. $f(x) = 3x^4 + 0 \cdot x^3 + 0 \cdot x^2 + (-1) \cdot x + 7$, so $a_4 = 3$, $a_3 = 0$, $a_2 = 0$, $a_1 = -1$, and $a_0 = 7$. Powers of x that do not appear have coefficient 0.
2. The highest power with a nonzero coefficient is x^4, so the degree is 4 and the leading coefficient is $a_4 = 3$.

Definition 0.1 (Constant function)**.** A function $f : A \to B$ is **constant** if there is some $c \in B$ with $f(x) = c$ for all $x \in A$.

Example 0.1 (Constant and non-constant functions)**.**

- $f(x) = 7$ on $\mathbb{R}$ is constant, with $c = 7$.
- $f(x) = x^2$ on $\mathbb{R}$ is not constant: $f(0) = 0$ but $f(1) = 1$.

Definition 0.2 (Polynomial)**.** A **polynomial** (in one variable x) [1] is a function $f : \mathbb{R} \to \mathbb{R}$ that can be written as

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

for some integer $n \geq 0$ and constants $a_0, a_1, \ldots, a_n \in \mathbb{R}$ with $a_n \neq 0$. The terms of f are $a_n x^n, \ldots, a_1 x, a_0$.

Remark (Constant and zero polynomials)

Remark 0.1 (Constant and zero polynomials). A constant function $f(x) = 7$ is a polynomial with $n = 0$ and $a_0 = 7$. The requirement $a_n \neq 0$ means this definition covers nonzero polynomials only: the zero function $f(x) = 0$ is excluded here, because it has no nonzero coefficient to serve as a_n.

Definition 0.3 (Coefficients of a polynomial)**.** The constants $a_0, a_1, \ldots, a_n$ of a polynomial f (Definition 0.2) are the **coefficients** of f, with a_k the coefficient of x^k.

Definition 0.4 (Degree of a polynomial)**.** Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with $a_n \neq 0$. The integer n is the **degree** of f.

Example 0.2 (Degrees of some polynomials)**.**

- $f(x) = 4x$ has degree 1.
- $f(x) = 5 - 2x^2 + x^3$ has degree 3: the degree is the highest power with a nonzero coefficient, not the power in the first term written.
- A constant polynomial $f(x) = 7$ has degree 0.

Definition 0.5 (Leading coefficient)**.** Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial of degree n. The constant a_n is the **leading coefficient** of f.

Example 0.3 (Leading coefficients)**.**

- $f(x) = 5 - 2x^2 + x^3$ has degree 3, so its leading coefficient is $a_3 = 1$, not the 5 written first.
- $f(x) = 3 - x^2$ has leading coefficient $a_2 = -1$.

Definition 0.6 (Quadratic polynomial)**.** A polynomial of degree 2, $f(x) = a_2x^2 + a_1x + a_0$ with $a_2 \neq 0$, is a **quadratic** [2] polynomial.

Definition 0.7 (Cubic polynomial)**.** A polynomial of degree 3 is a **cubic** polynomial.

Example 0.4 (Quadratic and cubic polynomials)**.**

- $$\begin{aligned} f(x) &= (x-2)^2 \\ &= x^2 - 4x + 4 \end{aligned}$$

  is quadratic, by Remark 5 in Convexity, Infimum and Sums with $a = x$ and $b = 2$.
- $f(x) = x^3 - 3x$ is cubic.
- $f(x) = 4x + 1$ is neither: it has degree 1.

Definition 0.8 (Parabola)**.** A **parabola** [3] is the graph of a quadratic polynomial (Definition 0.6).

Example 0.5 (The parabola $y = x^2$)**.** The graph of $f(x) = x^2$ is a parabola. It contains the points $(-1, 1)$, $(0, 0)$ and $(2, 4)$, and its lowest point is $(0, 0)$, since $x^2 \geq 0$ for every x.

Affine and linear functions of one variable

Definition 0.9 (Affine function of one variable)**.** A function $f : \mathbb{R} \to \mathbb{R}$ is **affine** [4] if there are numbers m and b such that

$$f(x) = mx + b \quad \text{for all } x \in \mathbb{R}$$

Definition 0.10 (Slope)**.** In the formula $f(x) = mx + b$ for an affine function f (Definition 0.9), m is called the **slope** of f [5].

Definition 0.11 (Intercept)**.** In the formula $f(x) = mx + b$ for an affine function f (Definition 0.9), b is called the **intercept** of f.

Definition 0.12 (Linear function of one variable)**.** A function $f : \mathbb{R} \to \mathbb{R}$ is **linear** [6] if it is affine (Definition 0.9) with intercept 0; that is, if there is a number m such that

$$f(x) = mx \quad \text{for all } x \in \mathbb{R}$$

Theorem 0.1 (Intercept and slope of an affine function)**.** *Let $f(x) = mx + b$ be an affine function (Definition 0.9).*

1. *The intercept is the value at zero: $f(0) = b$.*
2. *The slope is the change in f per unit change in x: for any $x_1 \neq x_2$,*

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = m$$

3. *So f determines its slope and intercept: if also $f(x) = m'x + b'$ for all x, then $m' = m$ and $b' = b$.*

Proof

Proof. **Part 1.**

$$\begin{aligned} f(0) &= m \cdot 0 + b && \text{(Definition 0.9)} \\ &= 0 + b && \text{(any number times 0 is 0)} \\ &= b + 0 && \text{(Theorem 5 in Convexity, Infimum and Sums)} \\ &= b && \text{(Theorem 4 in Convexity, Infimum and Sums)} \end{aligned}$$

Part 2. First the numerator:

$$\begin{aligned} f(x_2) - f(x_1) &= (mx_2 + b) - (mx_1 + b) && \text{(Definition 0.9)} \\ &= mx_2 + b - mx_1 - b && \text{(subtracting a sum subtracts each term)} \\ &= mx_2 - mx_1 + b - b && \text{(Theorem 5 in Convexity, Infimum and Sums, swapping } b \text{ and } -mx \\ &= (mx_2 - mx_1) + (b - b) && \text{(Theorem 6 in Convexity, Infimum and Sums)} \\ &= (mx_2 - mx_1) + 0 && (b - b = 0) \\ &= mx_2 - mx_1 && \text{(Theorem 4 in Convexity, Infimum and Sums)} \\ &= mx_2 + m(-x_1) && (-(mx_1) = m(-x_1), \text{ by Theorem 6 in Equalities, Inequalities and I} \\ &= m(x_2 + (-x_1)) && \text{(Theorem 11 in Convexity, Infimum and Sums, read right to left)} \\ &= m(x_2 - x_1) && \text{(adding a negative is subtracting)} \end{aligned}$$

Then, writing $d = x_2 - x_1$, which is not 0 because $x_1 \neq x_2$,

$$\begin{aligned}
\frac{f(x_2)-f(x_1)}{x_2-x_1} &= \frac{md}{d} && \text{(the numerator above)}\\
&= (md)\cdot\frac{1}{d} && \text{(Theorem 10 in Convexity, Infimum and Sums)}\\
&= m\cdot\left(d\cdot\frac{1}{d}\right) && \text{(Theorem 9 in Convexity, Infimum and Sums)}\\
&= m\cdot\frac{d}{d} && \text{(Theorem 10 in Convexity, Infimum and Sums)}\\
&= m\cdot 1 && \text{(a nonzero number divided by itself is 1)}\\
&= m && \text{(Theorem 7 in Convexity, Infimum and Sums)}
\end{aligned}$$

Part 3. Part 1, applied to each formula, gives

$$\begin{aligned} b &= f(0)\\ &= b'. \end{aligned}$$

Part 2 with $x_1 = 0$ and $x_2 = 1$, applied to each formula, gives

$$\begin{aligned} m &= f(1) - f(0)\\ &= m'. \end{aligned}$$

□

Example 0.6 (An affine function, and a linear one)**.** $f(x) = 2x + 3$ is affine with slope 2 and intercept 3. By Theorem 0.1, $f(0) = 3$, and each unit step in x raises f by 2: for example, $f(1) = 5$ and $f(2) = 7$. Its intercept is not 0, so it is not linear.
$g(x) = -\frac{1}{2}x$ is affine with intercept 0, so it is linear: $g(0) = 0$, and each unit step in x lowers g by $\frac{1}{2}$.

ℹ Remark (Elementary algebra calls $mx + b$ "linear")

Remark 0.2 (Elementary algebra calls $mx + b$ "linear")*.* Elementary algebra usually calls $f(x) = mx + b$ a *linear function*, because its graph is a straight line. This site calls that an affine function and keeps "linear" for the case $b = 0$, because that is the sense used in linear algebra, where a linear map must send 0 to 0 (theorem).

Limits of sequences

Definition 0.13 (Limit of a sequence)**.** A real number L is the **limit** of a sequence (a_n) of real numbers [7], written $\lim_{n\to\infty} a_n = L$ or $a_n \to L$, if for every $\varepsilon > 0$ there is a natural number N such that

$$|a_n - L| < \varepsilon \quad \text{for every } n \geq N.$$

Example 0.7 (The limit of a sequence)**.** $a_n = \frac{1}{n}$ has limit 0. Given $\varepsilon > 0$, take N to be any natural number larger than $\frac{1}{\varepsilon}$. For every $n \geq N$, $\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. For example, with $\varepsilon = 0.01$, take $N = 101$: every $n \geq 101$ has $\frac{1}{n} \leq \frac{1}{101} < 0.01$.

Definition 0.14 (Convergent sequence (converges))**.** A sequence (a_n) of real numbers **converges** to a real number L if L is its limit (Definition 0.13). A sequence that converges to some real number is **convergent**.

Example 0.8 (A convergent sequence)**.** $a_n = \frac{1}{n}$ converges to 0, because 0 is its limit (Example 0.7), so $a_n = \frac{1}{n}$ is convergent.

Definition 0.15 (Divergent sequence (diverges))**.** A sequence that does not converge (Definition 0.14) to any real number **diverges**, and a sequence that diverges is **divergent**.

Example 0.9 (A divergent sequence)**.** $c_n = n$ diverges. For any real number L and $\varepsilon = 1$, every $n > L + 1$ has $|n - L| > 1$, so no N works. For example, with $L = 5$, every $n \geq 7$ has $|n - 5| \geq 2$. It diverges to ∞: for every real number M, every $n \geq N$ has $c_n = n > M$ when N is a natural number larger than M.

Definition 0.16 (Divergence of a sequence to infinity)**.** A sequence (a_n) of real numbers **diverges to** ∞, written $\lim_{n\to\infty} a_n = \infty$, if for every real number M there is a natural number N such that $a_n > M$ for every $n \geq N$.

Series

Definition 0.17 (Non-decreasing sequence)**.** A sequence $a_1, a_2, \ldots$ of real numbers or of extended non-negative reals is **non-decreasing** if each term is at least as large as the one before it:

$$a_n \leq a_{n+1} \quad \text{for every } n \in \mathbb{N}$$

Example 0.10 (Non-decreasing and not)**.**

- $1, 2, 2, 3, 3, 3, \ldots$ is non-decreasing: a term may equal the one before it.
- $1, 2, \infty, \infty, \ldots$ is non-decreasing, because ∞ is greater than every real number.
- $1, \frac{1}{2}, \frac{1}{3}, \ldots$ is not non-decreasing, because

$$\begin{aligned} a_2 &= \frac{1}{2} \\ &< 1 \\ &= a_1. \end{aligned}$$

Definition 0.18 (Partial sum)**.** Let $a_1, a_2, \ldots$ be a sequence of real numbers or of extended non-negative reals. For each $n \in \mathbb{N}$, its nth **partial sum** [8] is the sum of its first n terms:

$$s_n \stackrel{\text{def}}{=} \sum_{i=1}^{n} a_i = a_1 + \cdots + a_n$$

Example 0.11 (Partial sums of halves)**.** For $a_i = 1/2^i$, the sequence $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$, the first three partial sums are $s_1 = \frac{1}{2}$,

$$\begin{aligned} s_2 &= \frac{1}{2} + \frac{1}{4} \\ &= \frac{3}{4}, \end{aligned}$$

and

$$\begin{aligned} s_3 &= \frac{3}{4} + \frac{1}{8} \\ &= \frac{7}{8}. \end{aligned}$$

In general, $s_n = 1 - 1/2^n$.

Definition 0.19 (Infinite sum, series)**.** Let $a_1, a_2, \ldots$ be a sequence in $[0, \infty]$, with partial sums s_n (Definition 0.18). The **infinite sum** (also called a **series**) [8] $\sum_{i=1}^{\infty} a_i$ is the limit of the partial sums:

$$\sum_{i=1}^{\infty} a_i \stackrel{\text{def}}{=} \lim_{n \to \infty} s_n$$

Here the limit is a real number when the partial sums are real numbers that converge to it, and it is ∞ when the partial sums diverge to ∞ or some partial sum is ∞. Sums with other index ranges, such as $\sum_{k=0}^{\infty} a_k$, are defined the same way.

Example 0.12 (A finite and an infinite sum)**.**

- For $a_i = 1/2^i$, the partial sums $s_n = 1 - 1/2^n$ (Example 0.11) converge to 1, because $1/2^n \to 0$. So $\sum_{i=1}^{\infty} 1/2^i = 1$.
- For $a_i = 1$, the partial sums are $s_n = n$, which diverge to ∞. So $\sum_{i=1}^{\infty} 1 = \infty$.

Lemma 0.1 (Sums of non-negative terms)**.** *Let $a_1, a_2, \ldots$ be values in $[0, \infty]$, with partial sums s_n (Definition 0.18), where $x + \infty = \infty$ for every x in $[0, \infty]$. Then the partial sums are non-decreasing, $s_1 \le s_2 \le \cdots$, so their limit, the infinite sum $\sum_{i=1}^{\infty} a_i$ (Definition 0.19), always exists, and it is either a finite number or ∞.*

Proof

Proof. For each n:

$$\begin{aligned} s_{n+1} &= s_n + a_{n+1} && \text{(definition of } s_{n+1}\text{)} \\ &\ge s_n && (a_{n+1} \ge 0) \end{aligned}$$

If some partial sum s_N is ∞, then $s_n = \infty$ for every $n \ge N$, so the limit is ∞. Otherwise, the partial sums form a non-decreasing sequence of real numbers. If that sequence is bounded above, it converges to a finite number [9]. If it is not bounded above, then for every number M some s_N exceeds M, and so does every later $s_n \ge s_N$; the limit is ∞. □

Exponentials and Logarithms

Definition 0.20 (Exponential function)**.** The **exponential function** [10] $\exp : \mathbb{R} \to (0, \infty)$ is

$$\exp\{x\} \stackrel{\text{def}}{=} \lim_{n \to \infty} \left(1 + \frac{x}{n}\right)^n,$$

the limit of the sequence $(1 + \frac{x}{1})^1, (1 + \frac{x}{2})^2, (1 + \frac{x}{3})^3, \ldots$. That limit exists for every real number x, and it is positive.

Example 0.13 (Approximating $\exp\{1\}$)**.** For $x = 1$, the terms $(1 + \frac{1}{n})^n$ are:

n	$(1 + \frac{1}{n})^n$
1	2
10	2.59374 ...
100	2.70481 ...
1000	2.71692 ...

They approach $\exp\{1\} = 2.71828 \ldots$. For $x = 0$, every term is $(1 + 0)^n = 1$, so $\exp\{0\} = 1$.

Definition 0.21 (Euler's number)**. Euler's number** [11] is

$$e \stackrel{\text{def}}{=} \exp\{1\} = 2.71828 \ldots$$

Example 0.14 (Euler's number is irrational)**.** $e = 2.71828 \ldots$ is an irrational number, so no fraction equals it exactly. The fraction $\frac{19}{7} = 2.714 \ldots$ is close, but $\frac{19}{7} \neq e$ [11].

Definition 0.22 (Natural logarithm)**.** For a real number $a > 0$, the **natural logarithm** of a, written $\log\{a\}$ [12], is the real number y with $\exp\{y\} = a$. The exponential function takes each positive value at exactly one input, so there is exactly one such y, and $\log : (0, \infty) \to \mathbb{R}$ is the inverse function of exp (Definition 0.20).

i Remark (Other notations for the natural logarithm)

Remark 0.3 (Other notations for the natural logarithm)*.* In these notes, $\log\{x\}$ is always the natural logarithm, the logarithm with base e (Definition 0.21) [13]. Some sources write $\ln x$ for the natural logarithm and reserve $\log x$ for the logarithm with base 10.

Example 0.15 (Natural logarithms)**.**

- $\log\{1\} = 0$, because $\exp\{0\} = 1$ (Example 0.13).
- $\log\{e\} = 1$, because $\exp\{1\} = e$ (Definition 0.21).
- $\log\{0\}$ and $\log\{-2\}$ are not defined, because $\exp\{y\} > 0$ for every real y.

Definition 0.23 (Power with a real exponent)**.** For a real number $a > 0$ and a real number b,

$$a^b \stackrel{\text{def}}{=} \exp\{b \cdot \log\{a\}\}.$$

When b is an integer, this agrees with Definition 5 in Equalities, Inequalities and Minimizers and Definition 9 in Equalities, Inequalities and Minimizers.

Example 0.16 (Real powers, and e^x)**.**

- $2^3 = \exp\{3 \cdot \log\{2\}\} \approx \exp\{3 \cdot 0.69315\} \approx \exp\{2.07944\} \approx 8$, which agrees with

$$\begin{aligned} 2^3 &= 2 \cdot 2 \cdot 2 \\ &= 8 \end{aligned}$$

  from Definition 5 in Equalities, Inequalities and Minimizers.

- $2^{1/2} = \exp\{\frac{1}{2} \cdot \log\{2\}\} \approx \exp\{0.34657\} \approx 1.41421$, which is $\sqrt{2}$ (Definition 10 in Equalities, Inequalities and Minimizers).

- For every real x, with base e (Definition 0.21):

$$\begin{aligned} e^x &= \exp\{x \cdot \log\{e\}\} && \text{(Definition 0.23, with } a = e \text{ and } b = x\text{)} \\ &= \exp\{x \cdot 1\} && \text{(}\log\{e\} = 1\text{, by Example 0.15)} \\ &= \exp\{x\} && \text{(Theorem 7 in Convexity, Infimum and Sums)} \end{aligned}$$

  So e^x and $\exp\{x\}$ are two names for the same number.

Definition 0.24 (Real powers of zero and of negative numbers)**.** Definition 5 in Equalities, Inequalities and Minimizers, Definition 9 in Equalities, Inequalities and Minimizers and Definition 0.23 leave some powers b^c with $b \leq 0$ unassigned. We complete them as follows.

- For a real number $c > 0$, $0^c \stackrel{\text{def}}{=} 0$. For a natural number c, this agrees with Definition 5 in Equalities, Inequalities and Minimizers.
- For $b < 0$, b^c is defined only when c is an integer, by Definition 5 in Equalities, In-

equalities and Minimizers and Definition 9 in Equalities, Inequalities and Minimizers; for $b < 0$ and a non-integer c, b^c is undefined.

Example 0.17 (Powers of zero and of negative numbers)**.**

- $0^{1/2} = 0$, which agrees with $\sqrt{0} = 0$ (Definition 10 in Equalities, Inequalities and Minimizers).
- $(-8)^2 = 64$ and $(-8)^{-1} = -\frac{1}{8}$ are defined, since the exponents are integers.
- $(-8)^{1/3}$ is undefined, even though $(-2)^3 = -8$. R follows the same convention: `(-8)^(1/3)` is `NaN`.

Theorem 0.2 (exp and log are mutual inverses)**.**

1. *For every* $a > 0$*:* $\exp\{\log\{a\}\} = a$.
2. *For every* $a \in \mathbb{R}$*:* $\log\{\exp\{a\}\} = a$.

Theorem 0.3 (Logarithm of a product)**.** *If* $a > 0$ *and* $b > 0$*, then*

$$\log\{a \cdot b\} = \log\{a\} + \log\{b\}$$

Corollary 0.1 (Logarithm of a quotient)**.** *If* $a > 0$ *and* $b > 0$*, then*

$$\log\left\{\frac{a}{b}\right\} = \log\{a\} - \log\{b\}$$

Proof

Proof. Since $a > 0$ and $b > 0$, the quotient $\frac{a}{b}$ is positive, so Theorem 0.3 applies to the product $\frac{a}{b} \cdot b$:

$$\begin{aligned} \log\{a\} &= \log\left\{\frac{a}{b} \cdot b\right\} && (a = \tfrac{a}{b} \cdot b) \\ &= \log\left\{\frac{a}{b}\right\} + \log\{b\} && \text{(logarithm of a product)} \end{aligned}$$

The second step applies Theorem 0.3. Subtracting $\log\{b\}$ from both sides gives $\log\{a\} - \log\{b\} = \log\{\frac{a}{b}\}$. □

Theorem 0.4 (Logarithm of a power)**.** *If* $a > 0$ *and* $b \in \mathbb{R}$*, then*

$$\log\{a^b\} = b \cdot \log\{a\}$$

Theorem 0.5 (Exponential of a sum)**.** *The exponential of a sum is equal to the product of the exponentials of its terms:*

$$\exp\{a+b\} = \exp\{a\} \cdot \exp\{b\}$$

Example 0.18 (Exponential of a sum)**.** With $a = 2$ and $b = 3$, $\exp\{2+3\} = \exp\{5\} \approx 148.41$, and $\exp\{2\} \cdot \exp\{3\} \approx 7.3891 \cdot 20.0855 \approx 148.41$.

Corollary 0.2 (Exponential of a difference)**.** *The exponential of a difference is the exponential of the first term divided by the exponential of the second term:*

$$\exp\{a-b\} = \frac{\exp\{a\}}{\exp\{b\}}$$

Example 0.19 (Exponential of a difference)**.** With $a = 5$ and $b = 2$, $\exp\{5-2\} = \exp\{3\} \approx 20.09$, and $\frac{\exp\{5\}}{\exp\{2\}} \approx \frac{148.4132}{7.3891} \approx 20.09$.

Theorem 0.6 (Powers of 1 and first powers)**.** *For every $b \in \mathbb{R}$,*

$$1^b = 1,$$

and for every $a \in \mathbb{R}$,

$$a^1 = a.$$

Theorem 0.7 (Power of a sum)**.** *If $a > 0$ and $b, c \in \mathbb{R}$, then*

$$a^{b+c} = a^b \cdot a^c$$

Example 0.20 (Power of a sum)**.** With $a = 2$, $b = 3$, and $c = 4$,

$$\begin{aligned} 2^{3+4} &= 2^7 \\ &= 128, \end{aligned}$$

and

$$\begin{aligned} 2^3 \cdot 2^4 &= 8 \cdot 16 \\ &= 128. \end{aligned}$$

Theorem 0.8 (Power of a product)**.** *If $a, b > 0$ and $c \in \mathbb{R}$, then*

$$(ab)^c = a^c \cdot b^c$$

When c is a positive integer, the same identity holds for all $a, b \in \mathbb{R}$, because both sides are products of c copies of a and c copies of b, which can be regrouped by Theorem 8 in Convexity, Infimum and Sums and Theorem 9 in Convexity, Infimum and Sums.

Example 0.21 (Power of a product)**.** With $a = 2$, $b = 3$, and $c = 2$,

$$\begin{aligned} (2 \cdot 3)^2 &= 6^2 \\ &= 36, \end{aligned}$$

and

$$\begin{aligned} 2^2 \cdot 3^2 &= 4 \cdot 9 \\ &= 36. \end{aligned}$$

Theorem 0.9 (Power of a power)**.** *If $a > 0$ and $b, c \in \mathbb{R}$, then*

$$\begin{aligned} a^{bc} &= (a^b)^c \\ &= (a^c)^b \end{aligned}$$

Example 0.22 (A negative base)**.** With $a = -1$, $b = 2$, and $c = \frac{1}{2}$:

$$\begin{aligned} a^{bc} &= (-1)^{2 \cdot \frac{1}{2}} \\ &= (-1)^1 \\ &= -1 \end{aligned}$$

but

$$\begin{aligned} (a^b)^c &= ((-1)^2)^{\frac{1}{2}} \\ &= 1^{\frac{1}{2}} \\ &= 1 \end{aligned}$$

So $a^{bc} \neq (a^b)^c$ here, which is why Theorem 0.9 requires $a > 0$. The third expression, $(a^c)^b = \left((-1)^{\frac{1}{2}}\right)^2$, is not even a real number.

Corollary 0.3 (Natural exponential of a product)**.**

$$\begin{aligned}\exp\{ab\} &= (\exp\{a\})^b \\ &= (\exp\{b\})^a\end{aligned}$$

Remark (Tarski's high school identities)

Remark 0.4 (Tarski's high school identities). Restricted to positive integers, the following results are Tarski's eleven "high school" identities [14]:

- sums are symmetric and associative (Theorem 5 in Convexity, Infimum and Sums, Theorem 6 in Convexity, Infimum and Sums);
- multiplying by 1 changes nothing, products are symmetric and associative, and multiplication is distributive (Theorem 7 in Convexity, Infimum and Sums, Theorem 8 in Convexity, Infimum and Sums, Theorem 9 in Convexity, Infimum and Sums, Theorem 11 in Convexity, Infimum and Sums);
- $1^b = 1$, $a^1 = a$, and the power of a sum, of a product, and of a power (Theorem 0.6, Theorem 0.7, Theorem 0.8, Theorem 0.9).

Remark (Tarski's identities are not complete)

Remark 0.5 (Tarski's identities are not complete). Tarski asked whether every identity in $+$, $\times$, exponentiation and 1 that is true for all positive integers can be derived from the eleven identities in Remark 0.4. It cannot: Wilkie found an identity that is true for all positive integers but does not follow from them. So this list is a useful core, not a complete rulebook.

Definition 0.25 (Contour line)**.** For a function g of two real variables b and c and a number k, the **contour line** [15] of g at height k is the set of points where g equals k:

$$\{(b, c) : g(b, c) = k\}$$

Example 0.23 (Contour lines of a bowl)**.** For $g(b, c) = b^2 + c^2$, the contour line at height $k = 4$ is $\{(b, c) : b^2 + c^2 = 4\}$, the circle of radius 2 around $(0, 0)$: for example,

$$\begin{aligned}g(2, 0) &= 4 + 0 \\ &= 4\end{aligned}$$

and

$$g(0,-2) = 0 + 4$$
$$= 4.$$

The contour line at height $k = -1$ is empty, because $b^2 + c^2 \geq 0$ for all b and c.

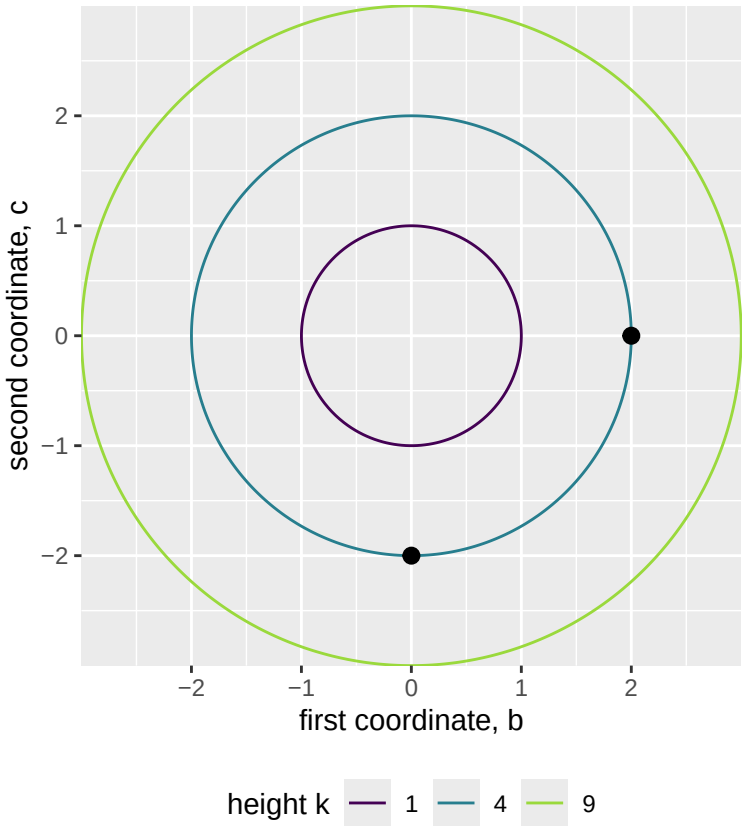


Figure 1: The contour lines of $g(b, c) = b^2 + c^2$ at heights 1, 4 and 9. The dots mark the points $(2, 0)$ and $(0, -2)$ of the example on the circle at height 4.

Remark (Contour lines on a plot of a surface)

Remark 0.6 (Contour lines on a plot of a surface). On a plot of the surface $z = g(b, c)$, the contour line is the curve along which the surface has height k.

Exercise 0.2 (Exponentiation versus multiplication)**.** For $b, c \in \mathbb{R}$, when does $b^c = bc$?

Solution

Solution 0.2. We only count a pair (b, c) when b^c is defined, so for $b < 0$ we only consider integer c (Definition 0.24). With that convention, $bc = b^c$ in each of the following cases:

1. $c = 1$, for every b.
2. $b = 0$ and $c > 0$, since then

$$\begin{aligned} b^c &= 0 \\ &= bc. \end{aligned}$$

   ($b = 0$ and $c = 0$ fails, since $0^0 = 1 \neq 0$.)

3. $b > 0$, $c > 0$, $c \neq 1$, and $b = \exp\left\{\frac{\log\{c\}}{c-1}\right\}$. For $b > 0$, dividing both sides of $b^c = bc$ by b gives $b^{c-1} = c$, which needs $c > 0$ because $b^{c-1} > 0$; taking logarithms then gives $(c-1)\log\{b\} = \log\{c\}$. For example, $c = 2$ gives $b = 2$, and indeed

$$\begin{aligned} 2^2 &= 4 \\ &= 2 \cdot 2. \end{aligned}$$

4. $b < 0$ and c is an odd integer with $c \geq 3$, with $b = -c^{1/(c-1)}$; for example, $b = -\sqrt{3}$ and $c = 3$ give

$$\begin{aligned} b^c &= -3\sqrt{3} \\ &= bc. \end{aligned}$$

5. $b < 0$ and c is an even integer with $c \leq -2$, with $b = -(-c)^{1/(c-1)}$; for example, $b = -2^{-1/3}$ and $c = -2$ give

$$\begin{aligned} b^c &= 2^{2/3} \\ &= bc. \end{aligned}$$

For $b < 0$, cases 4 and 5 come from $b^{c-1} = c$ as well: when $c - 1$ is even, $b^{c-1} > 0$, so c must be positive; when $c - 1$ is odd, $b^{c-1} < 0$, so c must be negative.
See the red contours in Figure 3 for a visualization of the $b \geq 0$ cases.

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 2: Graph of $b * c$ and b^c

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 3: **Graph of $b^c - b * c$.** The red contour lines are at heights 0 and 1; the ones at height 0 show where $b^c = b * c$.

Theorem 0.10 (When exponentiation equals multiplication)**.** *Let $b, c \in \mathbb{R}$, where b^c is defined (for $b < 0$, only integer c is allowed; Definition 0.24). Then $b^c = bc$ if and only if one of the following holds:*

1. *$c = 1$;*
2. *$b = 0$ and $c > 0$;*
3. *$b > 0$, $c > 0$, $c \neq 1$, and $b = \exp\left\{\frac{\log\{c\}}{c-1}\right\}$;*
4. *$b < 0$, c is an odd integer with $c \geq 3$, and $b = -c^{1/(c-1)}$;*
5. *$b < 0$, c is an even integer with $c \leq -2$, and $b = -(-c)^{1/(c-1)}$.*

Proof

Proof. This is the solution to Exercise 0.2, worked in Solution 0.2. □

Exercise 0.3 (Repeated exponentiation)**.** For $a \geq 0$, $b, c \in \mathbb{R}$, when does $(a^b)^c = a^{(b^c)}$?

Solution

Solution 0.3. Short answer: rarely (that's all you need to know for this course).
Long answer:
Split on whether $a > 0$ or $a = 0$, because the logarithm we use for $a > 0$ is undefined at $a = 0$.
Case $a > 0$. By Theorem 0.9, $(a^b)^c = a^{bc}$, so the question becomes when $a^{bc} = a^{(b^c)}$ (for pairs (b, c) where b^c is defined). Because $a > 0$, both sides are positive, and we can take logarithms (Theorem 0.4):

$$\begin{aligned} a^{bc} &= a^{(b^c)} \\ \log\{a^{bc}\} &= \log\{a^{(b^c)}\} \quad \text{(take logarithms of both sides)} \\ bc \cdot \log\{a\} &= b^c \cdot \log\{a\} \quad \text{(logarithm of a power)} \end{aligned} \tag{1}$$

The last line of Equation 1 holds exactly when

1. $a = 1$ (so that $\log\{a\} = 0$), or
2. $bc = b^c$ (see Theorem 0.10).

Case $a = 0$. Here we cannot take logarithms, so we work from the values of powers of 0: $0^s = 0$ for $s > 0$ (Definition 0.24), $0^0 = 1$, and 0^s is undefined for $s < 0$.

- If $b < 0$, then 0^b is undefined, so $(a^b)^c$ is undefined.
- If $b > 0$, then $(0^b)^c = 0^c$ and $b^c > 0$, so $0^{(b^c)} = 0$; the two sides agree exactly when $c > 0$.
- If $b = 0$, then

$$\begin{aligned}(0^0)^c &= 1^c \\ &= 1;\end{aligned}$$

for $c > 0$,

$$\begin{aligned}0^{(0^c)} &= 0^0 \\ &= 1,\end{aligned}$$

so the two sides agree, and for $c \leq 0$ they do not.

So for $a = 0$, $(a^b)^c = a^{(b^c)}$ exactly when $b \geq 0$ and $c > 0$.
In particular, when $a = 0$, $b \geq 0$, and $c = 0$, the two sides differ:

$$\begin{aligned}(a^b)^c &= (0^b)^0 \\ &= 1\end{aligned}$$

$$\begin{aligned}a^{(b^c)} &= 0^{(b^0)} \\ &= 0^1 \\ &= 0\end{aligned}$$

Theorem 0.11 (When repeated exponentiation is not reordered)**.** *Let $a \geq 0$ and $b, c \in \mathbb{R}$, where b^c is defined. Then $(a^b)^c = a^{(b^c)}$, with both sides defined, if and only if*

- *$a > 0$ and either $a = 1$ or $bc = b^c$; or*
- *$a = 0$, $b \geq 0$, and $c > 0$.*

Proof

Proof. This is the solution to Exercise 0.3, worked in Solution 0.3. □

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